

Evolution of Single Stars. VI. Model Nuclei of Planetary Nebulae

by

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ABSTRACT

Model evolutionary computations are presented for Population I ($X_0 = 0.7$, $Z = 0.03$) stars of 0.6 , 0.8 , and $1.2 M_\odot$. Each model consists of a degenerate carbon-oxygen core, helium and hydrogen burning shell sources and a low mass hydrogen rich envelope. These models evolve from right to left on the H—R diagram because matter flows from their envelopes through the shell sources into their cores. Superposed on this evolution there are loops caused by the thermal pulses within helium burning shell sources. Finally the available nuclear fuel becomes exhausted and models cool down to the white dwarf stage. Neutrino energy losses are included in the computations.

The initial model for every evolutionary sequence presented here was obtained from a model red supergiant stripped of most of its hydrogen rich envelope. These evolutionary sequences are believed to correspond to the evolutionary patterns of nuclei of planetary nebulae. The brightest and the youngest nuclei should be in a double shell source burning phase. Because of thermal pulses these stars are likely to heat up and cool down a number of times during the evolution. As a result a given planetary nebula may be ionized repeatedly during its lifetime.

1. Introduction

Nuclei of planetary nebulae are known to have luminosities of the order of $10^4 L_\odot$, and are known to evolve on a time scale of 10^4 years (O'Dell 1963, Seaton 1966). The observational evidence for these state-

ments has been reviewed by O'Dell (1968). It is generally agreed now that the ancestors of planetary nebulae and their nuclei are red supergiants, the idea due to Shklovsky (1956). The combined masses are believed to be close to $1 M_{\odot}$ or $1.5 M_{\odot}$, but very little is known about the dispersion of this quantity. A mechanism responsible for the ejection of a supergiant's envelope and creation of a planetary nebula is not known, although a large number of theories has been suggested (cf. *I.A.U. Symposium No. 34*). Also, a large variety of theoretical models has been proposed for the central stars of planetary nebulae. Few of these models were fitted to the known evolutionary history of a solar mass star. The only models that have been described so far, that have reasonable luminosities, and that have emerged from ordinary evolutionary computations are burning hydrogen and helium in the shell sources (Rose and Smith 1970, Paczyński 1970a = Paper I). This paper is devoted to a fairly detailed description of this kind of models. These results were already presented briefly in Paper I.

First of all let us try to identify the likely evolutionary status of nuclei of planetary nebulae. Let us assume that the neutrino energy losses due to the universal Fermi interactions exist. Under such circumstances a star with a mass below $1.4 M_{\odot}$ cannot ignite carbon at all, and a star with a mass in the range of $1.4 - 8 M_{\odot}$ ignites carbon explosively (Paczyński 1970a, 1971). This means that nuclei of planetary nebulae cannot be in a carbon burning phase. The only possibility for a low or moderate mass star to achieve a luminosity close to $10^4 L_{\odot}$ is to be in a double shell source burning phase. Prior to that stage the highest luminosity is reached just prior to the helium flash, with a hydrogen burning in a shell source. Even that luminosity is by an order of magnitude too low for a central star of planetary nebula. The phases of helium or hydrogen burning in the core are even less luminous. The models in the phase of gravitational contraction of the core following helium exhaustion are about as luminous as those burning helium in the core (Paczyński 1971, Table 1). Therefore we are left with models burning hydrogen and helium in the shell sources surrounding the degenerate carbon-oxygen core. In this evolutionary phase the luminosity may reach $5 \times 10^4 L_{\odot}$ (Paczyński 1970a, 1971 = Paper V).

How reliable are the luminosities of models with the two shell sources? It is well known that the helium burning shell source is thermally unstable, and it is technically impossible to follow the model evolution with hundreds of thermal pulses. Fortunately, simplified double shell source models have been computed with a variety of techniques. Their luminosities are plotted as a function of core mass

in Fig. 1. There is a scatter of results obtained by different authors, but it is clear that the desired luminosities are easily reached. Let us adopt the following core mass-stellar luminosity relation: $L/L_{\odot} = 59250 (M_{\text{core}}/M_{\odot} - 0.522)$, shown with a straight line in Fig. 1.

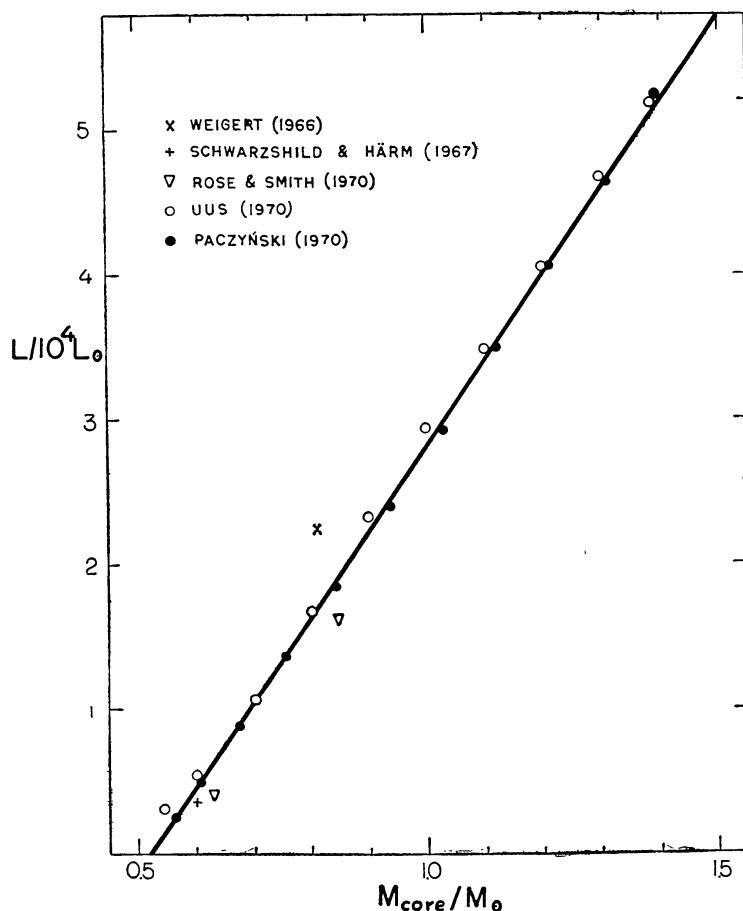


Fig. 1. Mass-luminosity relation for models with degenerate carbon-oxygen core and helium and hydrogen burning shell sources. The straight line shows the relation: $L/L_{\odot} = 59250 (M_{\text{core}}/M_{\odot} - 0.522)$.

Little is known about the mechanism that is responsible for the ejection of a planetary nebula. However, it seems that thermonuclear explosions may be excluded as there is no evolutionary phase known in a low or moderate mass star that could lead to a thermonuclear event on a dynamical time scale, in the case of explosion too high ejection velocities may be expected, and there is no reason to believe that either planetary nebulae or the atmospheres of their nuclei have unusual chemical composition expected in the case of thermonuclear outburst. The only evolutionary phase known in the moderate mass stars when one is likely to encounter a thermonuclear explosion is car-

bon ignition in the degenerate core. This phenomenon is on a supernova explosion scale (Arnett 1969), and may be excluded from our considerations. It is more likely that planetary nebula is ejected as a result of instability within the envelope or atmosphere of a red supergiant (Lucy 1967, Roxburgh 1967, Paczyński and Ziółkowski 1968*a*, *b*, Krishna Swamy and Stecher 1969, Finzi and Wolf 1970, 1971). Such an instability may be induced if the stellar luminosity is sufficiently high. It is interesting that at the time of a thermal pulse within the helium shell source the surface luminosity of the star does not change significantly (Schwarzschild and Härm 1967, Rose and Smith 1970), and may even decrease (Weigert 1966, this paper) as the hydrogen shell source becomes extinguished by the pulse. Therefore, there may be no obvious relation between the time when a thermal pulse appears in the helium shell, and the time when the envelope is lost.

Observations indicate that the youngest (*i.e.* the most compact) planetary nebulae have relatively cool central stars. The radii of those stars are two or three orders of magnitude larger than the radii of white dwarfs. It is clear that some part of the envelope has been left in such a star on the top of its degenerate core. There is no direct observational or theoretical information about the mass of that part of supergiant's envelope that is left after the major part of it has been ejected as a nebula. The nebula becomes visible not at the time of ejection, but when the central star becomes hot enough to ionize it. There may be a period when the nebula has already been ejected, and the central star is too cool to make it visible. It may also happen that the central star is already hot enough to ionize the nebula, and the mass outflow from the star is still going on, as it is observed in FG Sagittae (Herbig and Boyarchuk 1968) and in Wolf-Rayet type nuclei. Distinction should be made between the moment t_1 when the mass loss ends, and the moment t_2 when the nebula becomes visible. It may be that in some cases $t_1 > t_2$, and in others $t_1 < t_2$. Let us leave a discussion of the time scale of mass loss and the relation between t_1 and t_2 to a future paper. An approach to the problem of evolution of central stars that has been adopted in this paper is outlined below.

Let us assume that planetary nebulae are formed from ejected envelopes of red supergiants, and that those supergiants have degenerate carbon-oxygen cores surrounded by the helium and hydrogen burning shell sources. It is well known that the luminosity of such a model depends on the core mass only, while the stellar radius and effective temperature are influenced by the mass of hydrogen rich envelope (Paczyński 1970*a*, 1971, Uus 1970). Let us assume that at the time t_1 the mass loss has ended, and the subsequent evolution of the model

is not influenced by the total mass of the original supergiant or by the mass of ejected nebula. Let us follow the evolution of the remained star as if it was an isolated object. That star will evolve because the matter burning within the shell sources will flow from the envelope into the core. As a result the envelope will shrink and the effective temperature will increase. Sooner or later the shell sources will die out of lack of fuel, and the central star will cool down as a white dwarf. As long as the helium burning shell source remains active the picture outlined above will be complicated by the thermal instabilities within the shell source.

2. Simplified Models

Let us neglect for a moment all the complications in the evolution of a double shell source model due to the thermal pulses in the helium burning shell. The luminosity of such a model is determined by the core mass M_c . Let us add an envelope of the mass M_e to such a core. This will determine the effective temperature of the model. The total mass of the star is $M = M_c + M_e$. If the envelope mass is very small, *i.e.* $M_e \ll M_c$, we have approximately $M \approx M_c$, and we may use total mass — luminosity relation instead of core mass — luminosity relation. Let us adopt the relation shown with a straight line in Fig. 1: $L/L_\odot = 59250 (M/M_\odot - 0.522)$. Let us compute a series of static envelopes for a model with the mass M and the luminosity $L(M)$ for a range of effective temperatures. Almost all the envelope mass M_e is above the point where the temperature is 10^7 °K. Variation of the envelope mass with the effective temperature for the models of 0.6, 0.8, and $1.2 M_\odot$ is shown in Fig. 2. The luminosities of those models are consistent with their masses as long as $M \approx M_c$, *i.e.* $M_e \ll M$. All the envelopes have the composition: $X = 0.7$, $Z = 0.03$.

Evolution of a simplified model is due to the mass flow from the envelope through the shell sources into the core. The rate of mass flow is determined by the model luminosity and the energy content of 1 gram of the envelope matter. It is necessary to take into account energy release due to hydrogen and helium burning and to gravitational contraction. Carbon to oxygen ratio below the helium shell source and the gravitational energy depends on the core mass. Therefore the energy content of one gram of the envelope matter depends slightly on the core mass. All the necessary data for the models with „stationary” shell sources are given in Table 5 of Paper III (Paczynski 1970b). In the case of our models of 0.6, 0.8, and $1.2 M_\odot$ the luminosities are 4.8×10^3 , 1.61×10^4 ,

and $3.97 \times 10^4 L_{\odot}$ respectively, and the rates of mass flow are 6.0×10^{-8} , 2.0×10^{-7} , and $4.7 \times 10^{-7} M_{\odot}$ per year, respectively. Given the rate of mass flow and the relation between the envelope mass and the effective temperature (Fig. 2) it is possible to find the variation of the effective temperature with time. Our simplified models evolve horizontally on the H—R diagram from right to left. Evolutionary time scales given in Table 2 of Paper I were obtained with such simplified models for $\log T_e < 4.4$.

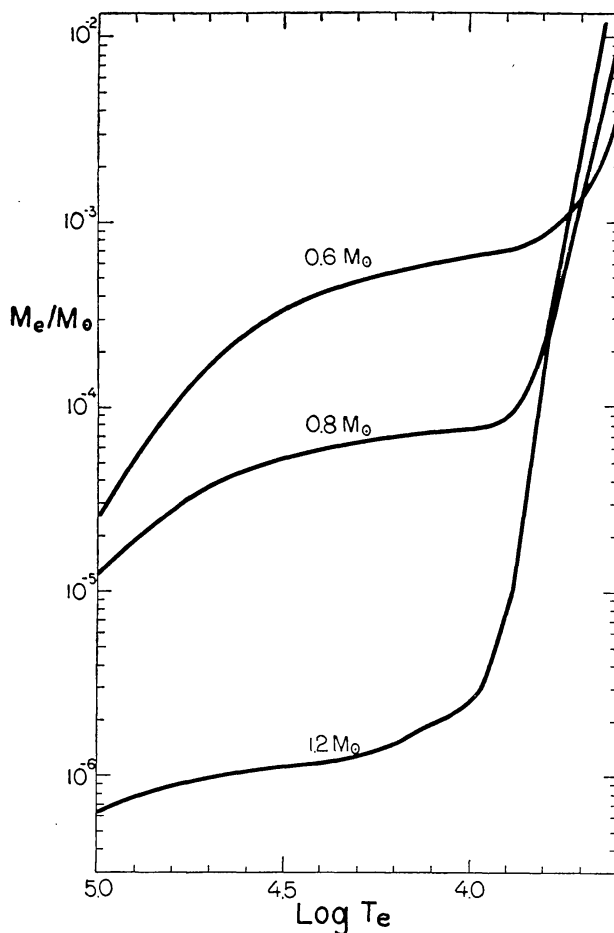


Fig. 2. Envelope mass as a function of effective temperature for the models of 0.6, 0.8, and 1.2 $M_{\odot}$. The luminosity for every model is adopted from the straight line relation shown in Fig. 1.

The lowest effective temperatures found among the nuclei of planetary nebulae are close to $\log T_e = 4.5$ (O'Dell 1968). Let us estimate the nuclear evolutionary time scale defined as the envelope mass corresponding to $\log T_e = 4.5$ divided by the rate of mass flow into the core. We get 5500, 260, and 2.3 years for the models of 0.6, 0.8, and

$1.2 M_{\odot}$, respectively. On this time scale all the nuclear fuel available in the envelope of a central star will be exhausted. 2.3 years is a remarkably short nuclear time scale indeed! Once this fuel is exhausted, the thermal cooling of the core becomes the dominant energy source, and our simplified models are no longer adequate.

Let us define a thermal time scale of the envelope as the total energy (gravitational + thermal) of that envelope divided by the luminosity of the model. It is surprising how little this quantity varies with the effective temperature, *i.e.* with the envelope mass of our simplified models. The thermal time scales for the envelopes of the 0.6 , 0.8 , and $1.2 M_{\odot}$ models are 4 years, 2.5 months and 1.5 days, respectively, for the following ranges of effective temperature: $\log T_e = 3.4 - 4.3$, $3.45 - 4.6$, and $3.7 - 4.9$, respectively. The thermal time scales are constant to within 20% for these ranges of effective temperature. Above the high temperature limit the time scale is getting shorter because of vanishingly small envelope mass. Below the low temperature limit the time scale is getting shorter as the total energy decreases. This phenomenon is caused by the large extent of the hy-

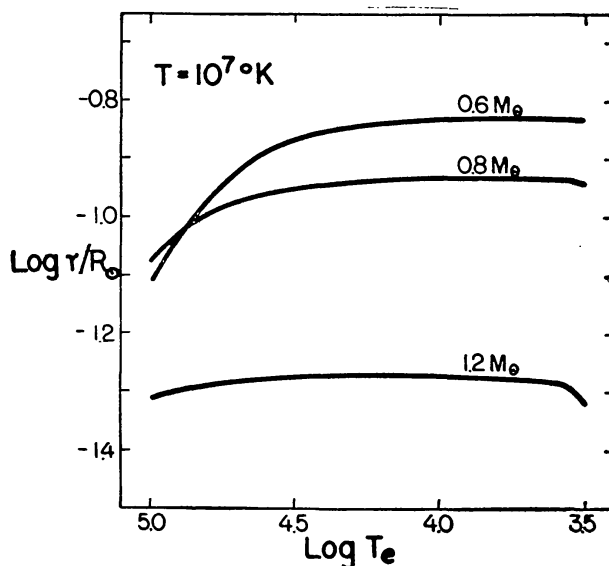


Fig. 3. Radii at the bottom of static envelopes ($T = 10^7$ °K) in the models of 0.6 , 0.8 , and $1.2 M_{\odot}$ as a function of effective temperature.

drogen and helium ionization zones and large recombination energies. At sufficiently low effective temperature the total energy changes sign and becomes positive (Paczynski 1969).

What is the range of envelope masses for which our simplified models are reasonable? Values of the density and radius were obtain-

ed at the point where the temperature was 10^7 °K, at the base of the envelopes. It turned out that the density was constant to within 0.1% for $3.7 < \log T_e < 5.0$, and that $\log \varrho = -1.2167$, -1.7922 , and -2.4785 in the 0.6 , 0.8 , and $1.2 M_\odot$ models, respectively. Variation of radius at $T = 10^7$ °K with the effective temperature of the model is shown in Fig. 3. This radius is reasonably constant over a fairly large range of $\log T_e$. At the high effective temperature the radius varies because the envelope mass is getting very small. At the low effective temperature the radius varies as the core mass is no longer „equal” to the total stellar mass, the envelope mass being too large. Within the range of effective temperatures where density and radius at $T = 10^7$ °K are constant the outer boundary conditions for the core are independent of the effective temperature, and the luminosity of the model should be constant. This is the region where the simplified models are reasonable. It is easy to extend this region to lower effective temperature and larger envelope masses by means of adjusting the model luminosity to the core mass rather than to the total mass of the star.

If the helium burning were stable the simplified models would be as good for the evolutionary studies as the complete models with the shell sources and the core explicitly included. Even with the instabilities of helium burning existing in nature but not taken into account here, the simplified models are realistic enough to give us an idea about the time scales involved, and the general direction of evolution of the central stars of planetary nebulae that may have two shell sources.

3. Evolutionary Computations

In order to follow in detail the evolution of a double shell source model, the shell instabilities, and the extinction of nuclear burning it is necessary to use a Henyey-type evolutionary code to integrate the time dependent stellar structure equations. In this chapter the results of model computations are presented for 0.6 , 0.8 , and $1.2 M_\odot$ stars.

It has been shown in Paper V (Paczynski 1971) that stars less massive than $8 M_\odot$ develop almost identical carbon-oxygen cores surrounded with the shell sources. The detailed core models ($0.6 < M_{\text{core}}/M_\odot < 1.4$) were obtained for a star of $3 M_\odot$. The cores of 0.6 , 0.8 , and $1.2 M_\odot$ were chosen from that sequence for this study. It should be pointed out that the original total masses of those three models might

have been quite different from $3 M_{\odot}$, however they should have been larger than their present masses, and smaller than $8 M_{\odot}$.

„Deep envelopes”, as described in Paper III (Paczynski 1970b), were fitted to the cores of 0.6 , 0.8 , and $1.2 M_{\odot}$, and such models were used as the initial conditions for the evolutionary computations. Logarithm of the effective temperature was arbitrarily chosen as 4.191 , 4.202 , and 4.728 for the three initial models. The corresponding mass

Table 1
Structure of the initial model of $0.6 M_{\odot}$.

$M_r/M_{\odot}$	$r/R_{\odot}$	$L_r/L_{\odot}$	$\log T$	$\log \varphi$	X	Y
0.000	0.0	0.0	7.985	6.322	0.0	0.0
0.070	3.9(-3)	-0.8	8.077	6.168	0.0	0.0
0.145	5.2	-2.0	8.148	6.046	0.0	0.0
0.246	6.7	-3.3	8.219	5.874	0.0	0.0
0.330	7.9	-2.4	8.249	5.704	0.0	0.0
0.391	8.8	-0.1	8.256	5.553	0.0	0.0
0.420	9.3	1.6	8.254	5.469	0.0	0.0
0.484	1.07(-2)	7.0	8.241	5.230	0.0	0.0
0.485	1.03	2.4	8.227	5.271	0.0	0.0
0.557	1.28	1.13(1)	8.210	4.691	0.0	0.001
0.578	1.47	1.22(2)	8.187	4.206	0.0	0.098
0.586	1.61	5.74	8.141	3.827	0.0	0.838
0.590	1.74	6.51	8.076	3.600	0.0	0.969
0.59871	2.79	6.72	7.725	1.706	0.001	0.969
0.59878	2.90	1.32(3)	7.718	1.467	0.110	0.860
0.59887	3.20	4.19	7.682	0.978	0.600	0.370
0.59901	4.07	4.77	7.573	0.589	0.699	0.271
0.59990	7.05(-1)	4.79	6.215	-3.727	0.700	0.270
0.59999	2.80(0)	4.79	5.429	-6.346	0.7	0.27
0.599999	5.59	4.79	4.953	-7.888	0.7	0.27
$(M-M_r)/M_{\odot}$						
9.6(-7)	5.87	4.79	4.928	-7.952	0.7	0.27
4.7	6.72	4.79	4.810	-8.387	0.7	0.27
1.2	8.15	5.79	4.593	-8.982	0.7	0.27
1.9(-8)	9.25	4.79	4.350	-9.629	0.7	0.27
3.2(-9)	9.86	4.79	4.182	-10.208	$\tau = 2/3$	
3.9(-10)	10.21	4.79	4.110	-10.726	$\tau = 0.05$	

of the region above the middle of the hydrogen burning shell source was equal to 1.2×10^{-3} , 1.4×10^{-4} , and $4 \times 10^{-6} M_{\odot}$ in those models. Every initial model consists of a dense degenerate core, a shell source region, and a static envelope. The structure of these three regions in the three initial models is presented in Tables 1—3. A Henyey-type code was used to follow the evolution of the shell source region in every model. A full set of time dependent stellar structure equations, subject to the boundary conditions, was integrated. The boundary conditions: $\partial \rho / \partial r = 0$, $\partial L_r / \partial t = 0$ were applied at the bottom of the shell source region. The first condition was reasonable as the inner boundary was always within the degenerate part of a star. The second condition was reasonable as long as the luminosity due to the gravitational contraction remained much larger than the luminosity at the inner boundary.

Table 2
Structure of the initial model of 0.8 M_⊙.

M _r /M _⊙	r/R _⊙	L _r /L _⊙	log T	log ϕ	X	Y
0.000	0.0	0.0	7.917	6.914	0.0	0.0
0.107	2.8(-3)	- 0.2	7.941	6.744	0.0	0.0
0.197	3.7	- 0.7	7.969	6.640	0.0	0.0
0.304	4.5	- 2.0	8.019	6.513	0.0	0.0
0.395	5.1	- 4.2	8.081	6.393	0.0	0.0
0.491	5.8	- 7.9	8.167	6.245	0.0	0.0
0.608	6.9	-11.2	8.276	5.998	0.0	0.0
0.669	7.5	- 7.9	8.310	5.814	0.0	0.0
0.670	7.3	- 3.2	8.318	5.846	0.0	0.0
0.790	1.05(-2)	6.19(1)	8.276	4.588	0.0	0.001
0.79582	1.14	3.57(2)	8.252	4.116	0.0	0.076
0.79782	1.22	1.80(3)	8.205	3.677	0.0	0.787
0.79847	1.27	2.14	8.155	3.477	0.0	0.967
0.79984	1.71	2.26	7.851	1.603	0.001	0.969
0.79985	1.76	4.36	7.844	1.332	0.108	0.862
0.79986	1.94	1.44(4)	7.802	0.783	0.620	0.350
0.79988	2.58	1.60	7.668	0.261	0.699	0.271
0.79995	1.02(-1)	1.61	7.010	- 1.799	0.700	0.270
0.79999	6.85	1.61	6.052	- 4.831	0.7	0.27
0.799999	4.72(0)	1.61	5.028	- 8.150	0.7	0.27
(M-M _r)/M _⊙						
9.9(-7)	5.22	1.61	4.974	- 8.322	0.7	0.27
4.5	8.94	1.61	4.698	- 9.270	0.7	0.27
1.8	12.85	1.61	4.510	- 9.844	0.7	0.27
4.6(-8)	16.27	1.61	4.282	-10.321	0.7	0.27
1.6	17.68	1.61	4.185	-10.595	τ = 2/3	
1.6(-9)	18.75	1.61	4.111	-10.914	τ = 0.05	

Table 3
Structure of the initial model of 1.2 M_⊙.

M _r /M _⊙	r/R _⊙	L _r /L _⊙	log T	log ϕ	X	Y
0.000	0.0	0.0	8.110	8.106	0.0	0.0
0.206	1.5(-3)	0.3	8.075	7.870	0.0	0.0
0.444	2.1	0.5	8.050	7.657	0.0	0.0
0.686	2.7	0.1	8.035	7.409	0.0	0.0
0.876	3.2	- 1.8	8.050	7.152	0.0	0.0
1.028	3.8	- 9.6	8.136	6.842	0.0	0.0
1.139	4.5	-30.6	8.348	6.394	0.0	0.0
1.175	4.9	-11.1	8.417	6.037	0.0	0.0
1.176	4.9	- 6.0	8.429	6.045	0.0	0.0
1.19927	5.9	2.09(2)	8.353	4.684	0.0	0.001
1.19980	6.1	1.18(3)	8.317	4.072	0.0	0.135
1.19991	6.3	4.55	8.268	3.654	0.0	0.856
1.19995	6.4	5.10	8.205	3.408	0.0	0.969
1.1999956	7.1	5.53	8.002	1.712	0.001	0.969
1.1999959	7.2	1.01(4)	7.997	1.447	0.098	0.872
1.1999964	7.6	3.40	7.964	0.774	0.598	0.372
1.1999971	9.4	3.91	7.845	0.257	0.699	0.271
1.1999986	2.08(-2)	3.95	7.407	- 1.217	0.700	0.270
1.1999991	4.70	3.96	7.002	- 2.507	0.7	0.27
1.1999998	9.50	3.97	6.674	- 3.603	0.7	0.27
1.1999998	1.91(-1)	3.97	6.302	- 4.683	0.7	0.27
(M-M _r)/M _⊙						
1.9(-7)	2.14	3.97	6.249	- 4.838	0.7	0.27
6.7(-8)	4.04	3.97	5.873	- 6.043	0.7	0.27
2.2	7.33	3.97	5.497	- 7.314	0.7	0.27
6.3(-9)	1.37(0)	3.97	5.132	- 8.494	0.7	0.27
4.0(-10)	2.48	3.97	4.704	- 9.851	τ = 2/3	
3.8(-11)	2.84	3.97	4.618	-10.640	τ = 0.05	

The outer boundary conditions for the shell source region were given in a table as a function of surface luminosity and temperature. The table was calculated by means of integrating a grid of static envelopes of the mass of $10^{-6} M_{\odot}$ for the $0.6 M_{\odot}$ and $0.8 M_{\odot}$ models, and the mass of $2 \times 10^{-7} M_{\odot}$ for the $1.2 M_{\odot}$ model. The assumption that the shell source region was stationary was used to get the initial model only, and has been abandoned later on. A hydrostatic equilibrium and a spherical symmetry was assumed throughout this study. Energy losses due to the universal Fermi interactions were taken into account.

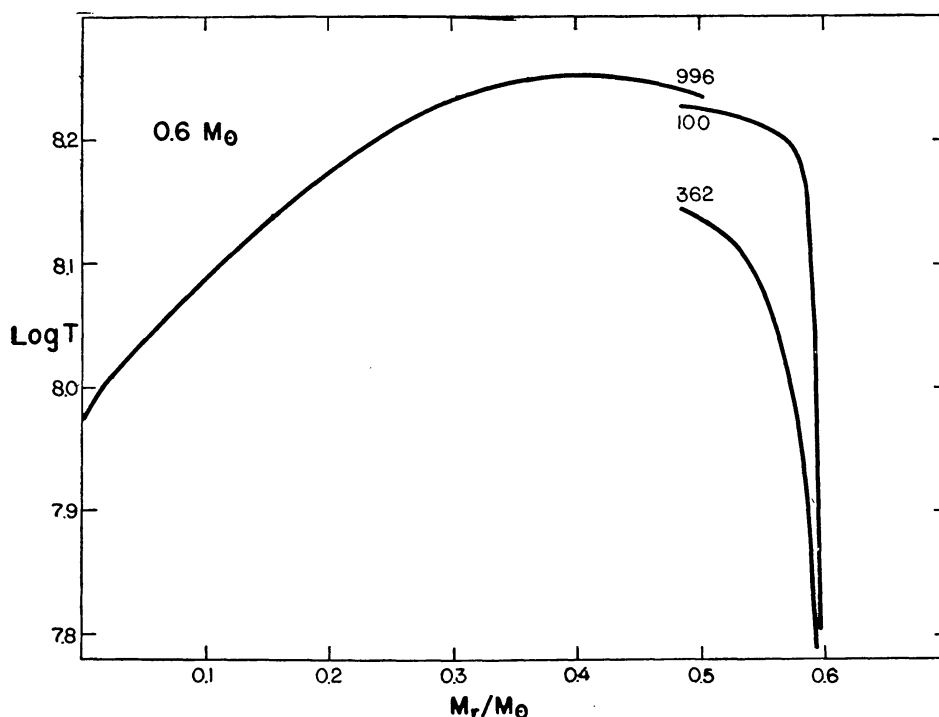


Fig. 4. Temperature distribution with mass in the model of $0.6 M_{\odot}$. The core model number 996 is taken from the evolutionary computations of the $3 M_{\odot}$ star (Paczynski 1971). The initial „deep envelope” (*i.e.* static envelope and a shell source region) has a number 100. The last computed model of the evolutionary sequence for the $0.6 M_{\odot}$ star has a number 362.

All the details of the equation of state, opacities, and nuclear reaction rates were identical with those described in Paper I. No mass loss was assumed from the models while their evolution was calculated. The chemical composition of the static envelopes was: $X = 0.7$, $Y = 0.27$, $Z = 0.03$. Evolution of the core was not studied.

Temperature distribution within the first and the last model of every evolutionary sequence is shown in Figs. 4–6. The first model in every sequence was given a number „100”. Evolution of the models

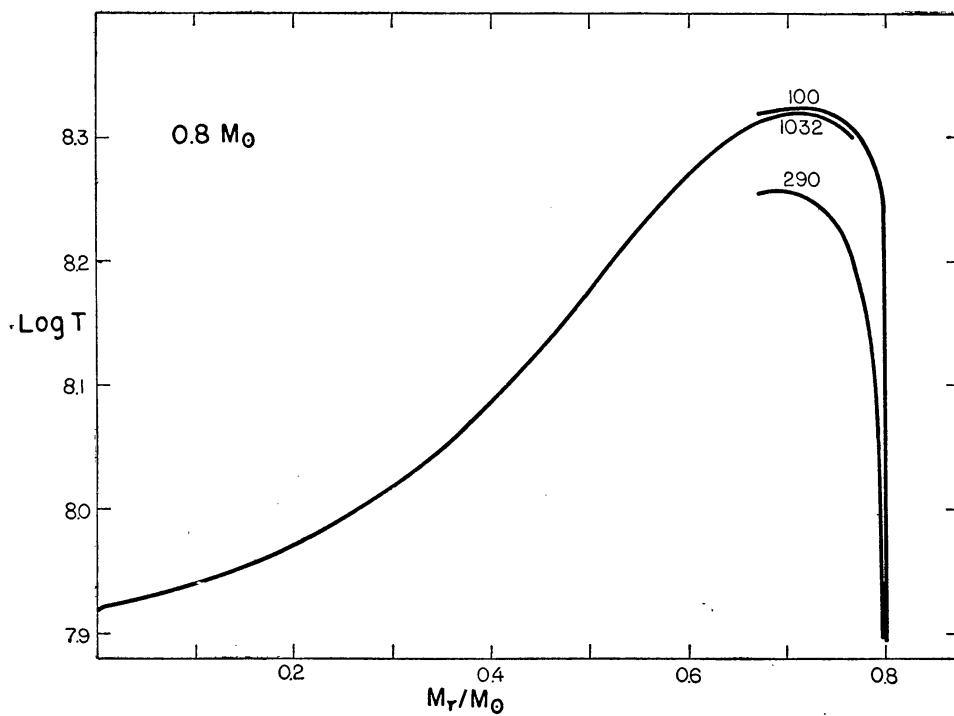


Fig. 5. The same as Fig. 4 but for the $0.8 M_\odot$ model.

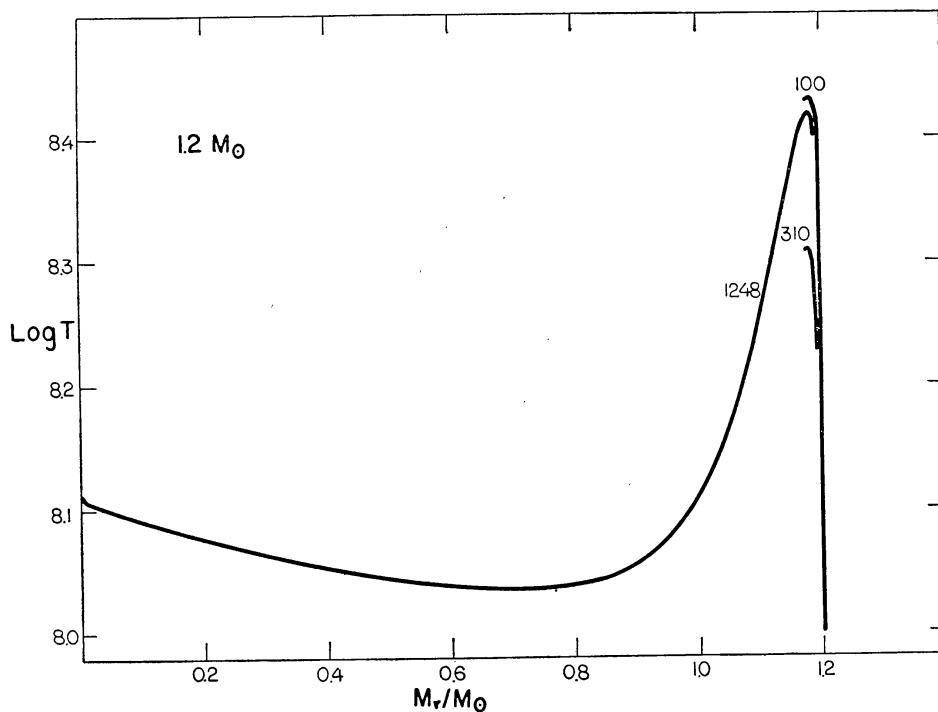


Fig. 6. The same as Fig. 4 but for the $1.2 M_\odot$ model.

in the $\log \varrho - \log T$ plane is presented in Figs. 7—9, and in the H—R diagram in Fig. 10. The last Figure is similar to Fig. 4 of Paper I, where the observed positions of central stars of planetary nebulae have been also shown. Evolutionary changes of some parameters of the three models are presented in Tables 4—6. L_H is the total luminosity (nuclear + gravitational) released within the hydrogen shell source and the envelope, L_{He} is the total luminosity released within helium shell

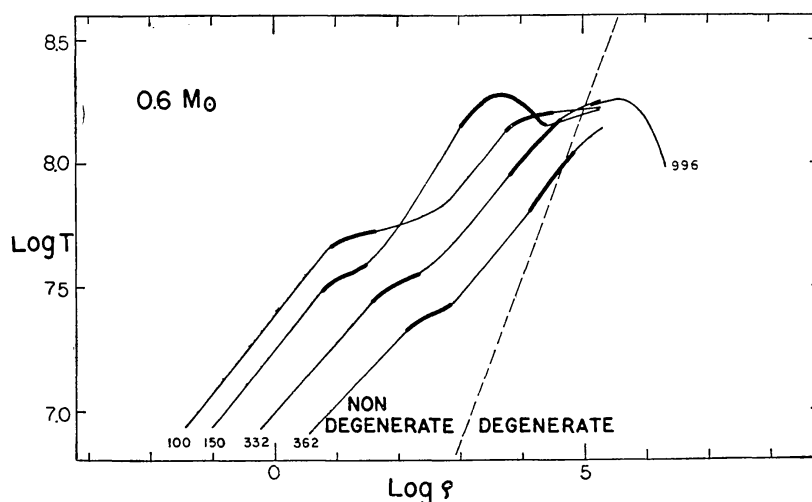


Fig. 7. Evolution of the shell source region in the $\log \varrho - \log T$ plane for the $0.6 M_{\odot}$ model. Every model is represented with a line labelled with the model number. Position of the helium burning shell source ($0.01 < Y < 0.90$) and the hydrogen burning shell source ($0.01 < X < 0.65$) is indicated with a thick section of the model line. Position of the initial carbon-oxygen core (number 996) is also shown.

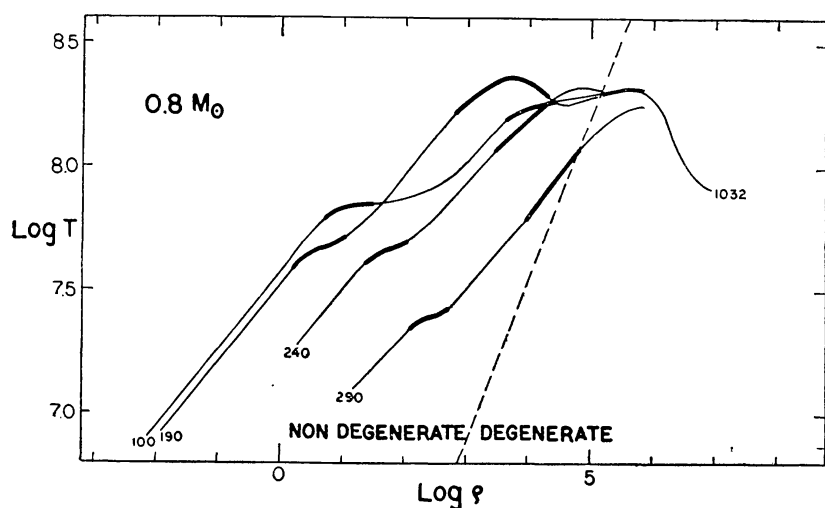


Fig. 8. The same as Fig. 7 but for the $0.8 M_{\odot}$ model.

source and the helium zone between the shell sources. L is the surface luminosity.

Given the initial conditions for the three models their farther evolution is believed to be realistic. Thermal instabilities develop in the helium shell sources in all three models, but they are not violent. In the $0.6 M_{\odot}$ star the maximum of the thermal pulse was reached in the model 145 when the highest luminosity within the star was 4.3 times

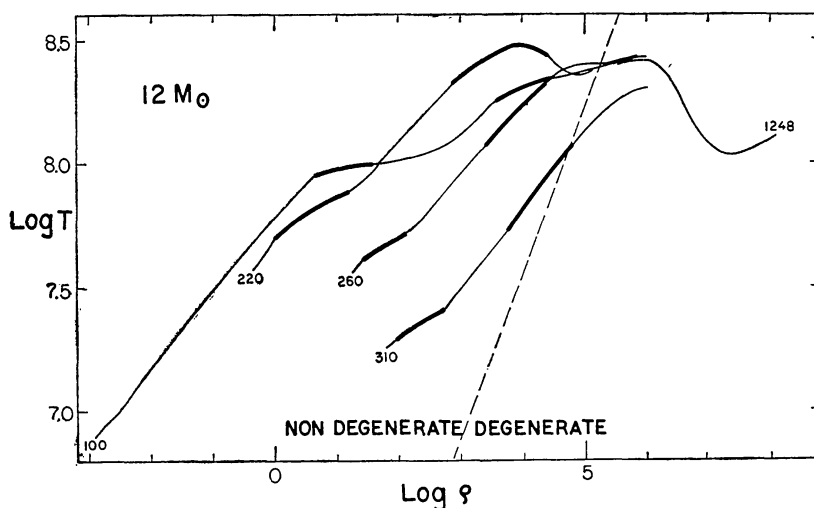


Fig. 9. The same as Fig. 7 but for the $1.2 M_{\odot}$ model.

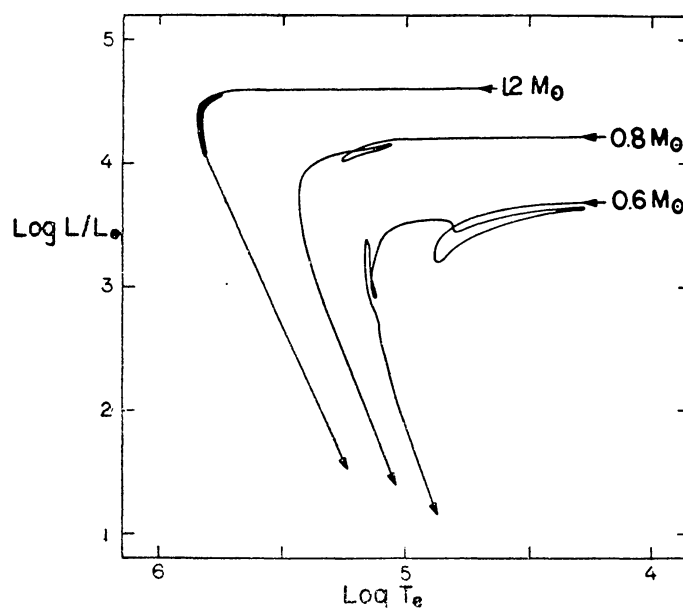


Fig. 10. Evolutionary tracks on the H—R diagram for the models of 0.6 , 0.8 , and $1.2 M_{\odot}$.

Table 4
Evolutionary changes of the 0.6 M_⊙ model.

Model	Time (years)	log T _e	L/L _⊙	L _H /L	L _{He} /L
100	0.16 (3)	4.191	4.79 (3)	0.861	0.138
110	1.38	4.390	4.66	0.851	0.147
120	2.04	4.606	4.18	0.820	0.177
130	2.09	4.680	3.90	0.804	0.192
140	2.15	4.848	2.68	0.669	0.324
150	2.31	4.833	1.79	-0.258	1.245
160	2.42	4.656	2.96	-0.164	1.159
170	2.50	4.477	3.71	-0.073	1.067
180	2.59	4.305	4.16	-0.019	1.012
190	2.84	4.430	4.03	0.029	0.962
200	3.06	4.595	3.58	0.034	0.959
210	3.83	4.737	2.96	0.033	0.982
220	1.04 (4)	4.820	3.26	0.594	0.428
230	1.49	4.895	3.46	0.734	0.269
240	1.82	4.982	3.30	0.779	0.219
250	2.11	5.066	2.76	0.784	0.210
260	2.44	5.126	1.73	0.723	0.263
270	2.83	5.147	1.47	0.759	0.220
280	2.94	5.144	1.07	0.677	0.292
290	3.04	5.153	1.27	0.749	0.224
300	3.07	5.166	1.84	0.835	0.147
310	3.13	5.134	1.05	0.706	0.261
320	3.35	5.114	4.71 (2)	0.417	0.506
332	4.94	5.066	1.94	0.257	0.550
342	8.50	5.008	7.99 (1)	0.192	0.456
352	1.58 (5)	4.942	3.33	0.128	0.409
362	3.03	4.868	1.39	0.078	0.371

larger than the surface luminosity. Convection appeared in the models 120—145, with the maximum extent from $Y=0.38$, $\log \varrho=3.53$ up to $Y=0.9697$, $\log \varrho=2.74$. In the 0.8 M_⊙ star the maximum of the pulse was reached in the model 180 with $L_{max}/L=1.6$. Convection was pre-

Table 5
Evolutionary changes of the 0.8 M_⊙ model.

Model	Time (years)	log T _e	L/L _⊙	L _H /L	L _{He} /L
100	0.10 (1)	4.202	1.61 (4)	0.861	0.136
110	0.85	4.240	1.62	0.859	0.137
120	1.12	4.249	1.61	0.857	0.139
130	2.59	4.350	1.61	0.859	0.138
140	5.55	4.559	1.61	0.858	0.138
150	1.09 (2)	4.777	1.59	0.851	0.146
160	1.69	4.967	1.55	0.822	0.174
170	1.99	5.126	1.41	0.746	0.250
180	2.23	5.255	1.05	0.149	0.844
190	2.63	5.068	1.34	-0.009	1.002
200	3.92	5.194	1.20	0.080	0.928
210	7.74	5.309	1.12	0.457	0.584
220	1.02 (3)	5.409	8.75 (3)	0.510	0.517
230	1.33	5.437	4.18	0.354	0.660
240	1.87	5.396	1.70	0.219	0.726
250	2.86	5.335	7.12 (2)	0.128	0.653
260	5.29	5.266	3.04	0.066	0.493
270	1.25 (4)	5.194	1.31	0.032	0.322
280	3.51	5.118	5.68 (1)	0.015	0.188
290	1.03 (5)	5.038	2.42	0.007	0.136

Table 6
Evolutionary changes of the $1.2\ M_{\odot}$ model.

Model	Time (years)	$\log T_e$	$L/L_{\odot}$	L_H/L	L_{He}/L
100	0.03	4.728	3.97 (4)	0.864	0.132
110	0.40	4.834	3.98	0.861	0.135
120	0.72	5.137	3.99	0.862	0.132
130	1.18	5.260	3.98	0.860	0.134
140	2.10	5.373	3.96	0.859	0.136
150	2.99	5.554	3.89	0.853	0.142
160	4.25	5.753	3.57	0.831	0.162
170	5.62	5.853	2.47	0.731	0.261
180	6.54	5.828	1.40	0.499	0.487
190	8.72	5.812	1.20	0.201	0.783
200	9.58	5.827	1.71	0.085	0.903
210	1.06 (1)	5.816	2.84	0.004	0.989
220	1.50	5.763	3.49	0.011	0.985
230	2.30	5.837	2.09	0.051	1.023
240	3.06	5.809	1.17	0.050	1.066
250	4.30	5.744	5.28 (3)	0.028	1.037
260	6.97	5.664	2.23	0.011	0.730
270	1.43 (2)	5.582	9.61 (2)	0.004	0.465
280	3.91	5.499	4.21	0.002	0.230
290	1.44 (3)	5.418	1.88	0.001	0.086
300	6.17	5.334	8.27 (1)	0.000	0.040
310	1.79 (4)	5.242	3.41	0.000	0.040

sent in models 170—175, with the maximum extent from $Y=0.57$, $\log \varrho=3.32$ up to $Y=0.968$, $\log \varrho=2.83$. In the $1.2\ M_{\odot}$ star the pulse was the strongest in the model 200 with $L_{max}/L=2.0$. Convection was present in the models 195—205 with the maximum extent from $Y=0.73$, $\log \varrho=3.35$ up to $Y=0.95$, $\log \varrho=3.03$. The strength of the thermal pulse decreases with the increasing core mass! Perhaps this surprising result is due to the smoothness of the initial, „stationary” profile of the chemical composition.

The mass of the intermediate helium zone between the two shell sources was 1.48×10^{-2} , 2.42×10^{-3} , and $1.3 \times 10^{-4}\ M_{\odot}$ in the first, „stationary” models of 0.6 , 0.8 , and $1.2\ M_{\odot}$, respectively. That mass was reduced to 1.4×10^{-2} , 2.0×10^{-3} , and $0.8 \times 10^{-4}\ M_{\odot}$, respectively, in the last models computed in every sequence. By that time the shell sources were no longer active. In those last models the mass of the hydrogen rich envelope above the helium region was reduced to 1.4×10^{-4} , 2.0×10^{-5} , and $0.75 \times 10^{-6}\ M_{\odot}$, respectively. As there was no nuclear burning within those models, the residual masses of the hydrogen and helium rich zones that were left close to the surface should remain constant when those models would cool down to the white dwarf stage.

All three stars evolved almost horizontally in the H—R diagram as long as the shell sources were active. This part of evolution may be qualitatively understood in terms of the simplified models presented in the previous chapter: a star is getting smaller and hotter because

the matter flows from the envelope into the core being burnt within the shell sources. However, superposed on this general evolutionary pattern there are the loops caused by the thermal instability of the helium shell source. Once the available nuclear fuel is exhausted, the shell sources die out, and a star cools down at a constant, white dwarf radius. It is interesting that the hydrogen shell source was always thermally stable. In the last models of the three evolutionary sequences the „hydrogen luminosity” L_H was negligible (cf. Tables 4—6), and no thermal instabilities are expected during the subsequent cooling of those stars.

4. Discussion

Comparison of the evolutionary tracks of the double shell source models with the observed positions of nuclei of planetary nebulae on the H—R diagram (Fig. 4 of Paper I) has shown that there is a qualitative agreement between the models and observations, the low luminosity, „old” models being too hot. The model evolutionary time scales are very sensitive to the stellar mass, but are of the same order of magnitude as the observed time scale. The bright „young” models with active shell sources evolve from right to left on the H—R diagram, on average. However, complicated loops due to the thermal pulses in the helium shell source are superposed on this „average” evolution. Every model presented in this paper went through only one thermal pulse. If the initial models with more massive envelopes were taken, a larger number of thermal pulses would appear prior to extinction of nuclear burning. Models with so massive envelopes would still be in the red supergiant region (cf. Fig. 2), and would be unable to excite the surrounding nebula. However, at the time of thermal runaway the surface luminosity and radius decrease in such a way that the effective temperature may increase significantly, provided the envelope mass is not too large. Such excursions to the high temperature region of the H—R diagram were observed by Schwarzschild and Härm (1970) in some of their double shell source models. If the effective temperature became sufficiently high, the nebula could be ionized. The model would return to the red supergiant region after the thermal pulse, and the nebula would recombine and disappear. This kind of activity might be repeated a number of times, as long as there is nuclear fuel available for thermal instabilities. A nebula may appear and disappear a number of times as a result of this activity of helium burning shell source. Perhaps FG Sagittae returns now to the red supergiant region after the thermal pulse in its interior.

It will be necessary to learn more about the process of mass loss from red and yellow supergiants, and about the formation of planetary nebulae before a real comparison between models and observations of central stars will be possible. Most likely the main driving force for the evolution of bright and young nuclei is the reduction of mass of their envelopes. That mass is burnt at the bottom and flows into the core, and is lost from the top and flows into the nebula. The models presented in this paper may account for the first of these processes only.

It is not possible at present to relate the mass of the nucleus to the original, total mass of the star on the basis of theoretical models only. If the envelopes of red supergiants were lost because of dynamical instability (Paczynski and Ziolkowski 1968b), the stars of 0.8 , 1.5 , and $3.0 M_{\odot}$ would produce the nuclei of 0.6 , 0.8 , and $1.2 M_{\odot}$, and the nebulae of 0.2 , 0.7 , and $1.8 M_{\odot}$. This relation is very uncertain. Observations of open clusters (masses at the turn off point and masses of the cluster white dwarfs) indicate that the relation given above is pretty good (Ostriker 1970, private communication). It is assumed here that all white dwarfs have evolved through the planetary nebula phase.

It is interesting that the model of $1.2 M_{\odot}$ presented in this paper evolved so fast that it could not ionize the whole nebula of $1.8 M_{\odot}$, even if such a nebula was around. The total number of ultraviolet quanta emitted by that model during the first 30 years, when $L/L_{\odot} > 10^4$, was sufficient to ionize only $0.44 M_{\odot}$ of the neutral gas. During the subsequent 100 years, when $10^4 > L/L_{\odot} > 10^3$, additional $0.18 M_{\odot}$ could be ionized. It is likely that massive planetary nebulae, even if they exist, are unobservable as massive because they are never fully ionized.

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